



Hochschule für
Wirtschaft und Recht Berlin
Berlin School of Economics and Law

Advanced Security Master Digital Transformation

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(Security and Embedded Systems Engineering)



Topics

Foundations

- Cryptography
- Authentication
- Authorization
- Network Intro & Security
- Web Application Security & Basic CTF
- Information Security Management & Operational Security Topics



Test: Foundations (50%)

Advanced Topics

- Embedded Security
- Cloud Security
- Ethical Hacking / Pentesting Intro
- Advanced Cryptography
- Machine Learning & Security
- Current Research Topics / Seminar

Final Exam: Security Use Cases in your businesses, presentation (50%)

Schedule



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April	May	June	July
1st presence: today (Apr 17)	Week 4th –8th async	4th presence	8th online
Week 20 th -24 th async	13th online	Week 8th –12th async	17th presence
Week Apr 27th- May 1st async	22nd online	19th online	
	Week 25th –29th async	Week 22nd – 26th async	
		Week 29th – July 3rd async	

Schedule



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April	May	June	July
Intro & Math Intro	Network Security	Q & A and test	Tell me what you want to deepen
Crypto	Web Application Security	Machine Learning & Security	Final Presentations
Authentication & Authorization	Q & A and OpSec	Q&A, selection of topics	
	Cloud Security	Embedded Security	
		Tell me what you want to deepen	



Capture The Flag (CTF)



- Perhaps possible if you like – need to check with the company

If you wanna join, let me know



Feedback for the plan

- What do you expect from the lecture?
- Any particular content you wish to focus at?
- Any content you are missing in the plan?
- Any content you wish to prioritise?
- Any content to down-prioritise?